

Econ 135: Lecture 10

Monetary Policy Implementation with a Large Central Bank Balance Sheet

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Outline

- The Federal Reserve's operating framework
- Monetary policy implementation before 2008
- Monetary policy implementation since 2008
 - The leaky floor
 - Why are banks holding so many excess reserves?
 - The future



Federal Open Market Committee (FOMC)

- The committee consists of
 - The seven members of the Federal Reserve Board
 - The president of the FRBNY
 - The presidents of four other Federal Reserve Banks
- Focal point for policy making in FRS
- The committee meets 8 times a year
- The FOMC issues directives to the trading desk at FRBNY



Monetary Policy Formulation and Implementation

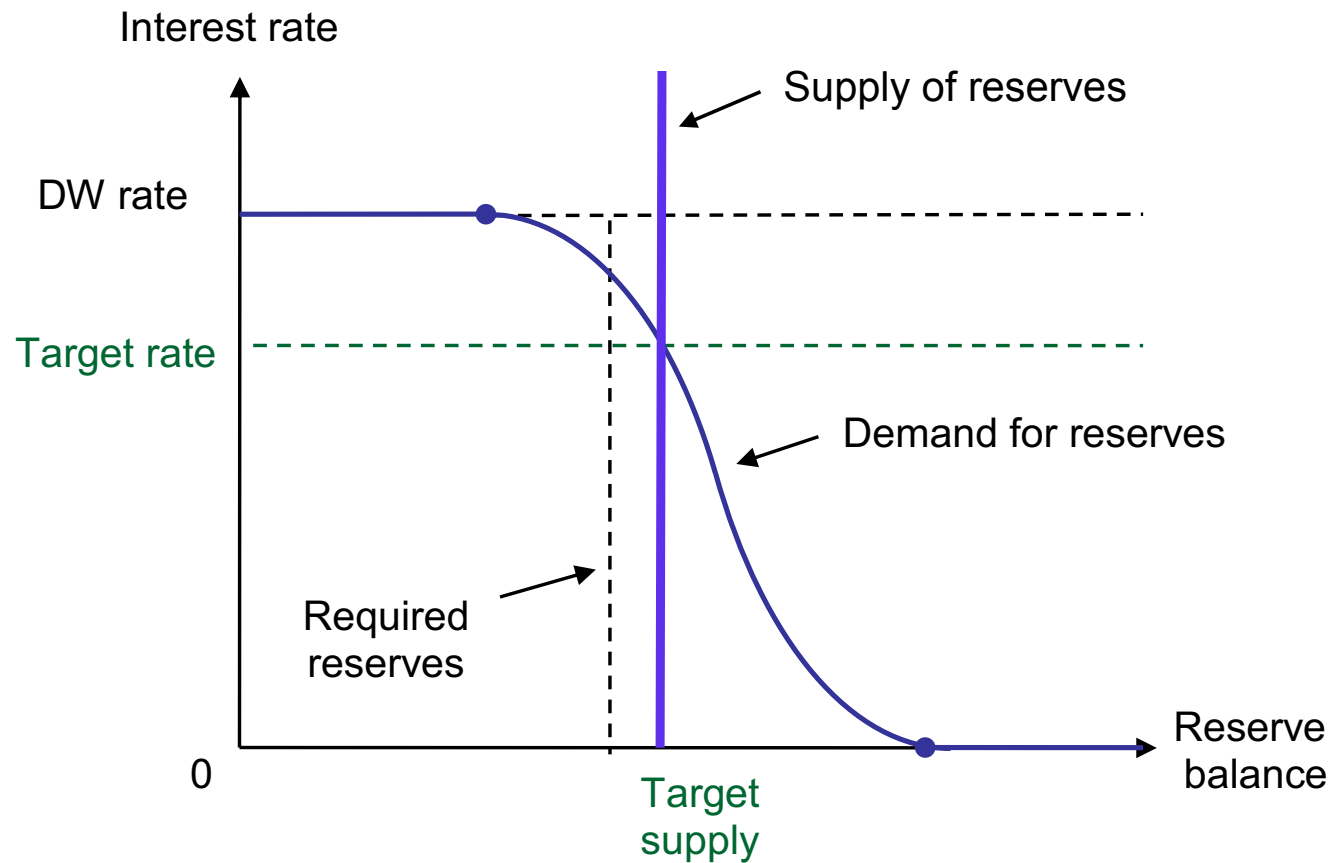
- Policy formulation
 - Objectives established by Federal Reserve Act
 - “Dual mandate” of maximum employment and price stability
- Policy implementation through open market operations (OMOs)
 - FOMC votes on domestic policy directive
 - Directive establishes an operating objective for OMOs
 - Desk designs and conducts OMOs to achieve the objective
 - SOMA Manager accountable to the FOMC
 - Market monitoring supports policy formulation and execution
- Board of Governors regulates complementary policy tools
 - Discount window (administered by Reserve Banks)
 - Reserve requirements, interest on reserves, term deposits

Operating Framework for Policy Implementation

- Conventional operating objective for OMOs was target federal funds rate (FFR)
 - Desk adjusted supply of reserve balances to achieve target
 - Repos/reverse repos address transitory reserve needs
 - Treasury purchases offset currency growth
- With the FFR at the effective zero bound, the operating objective shifted to adjustments in balance sheet size/composition
 - Large-scale purchases of Treasuries, agency MBS and debt
 - Interest on reserves keeps FFR in target range w/o OMOs
- FOMC has also used communications to influence policy

Monetary policy implementation before 2008

Monetary policy implementation before 2008

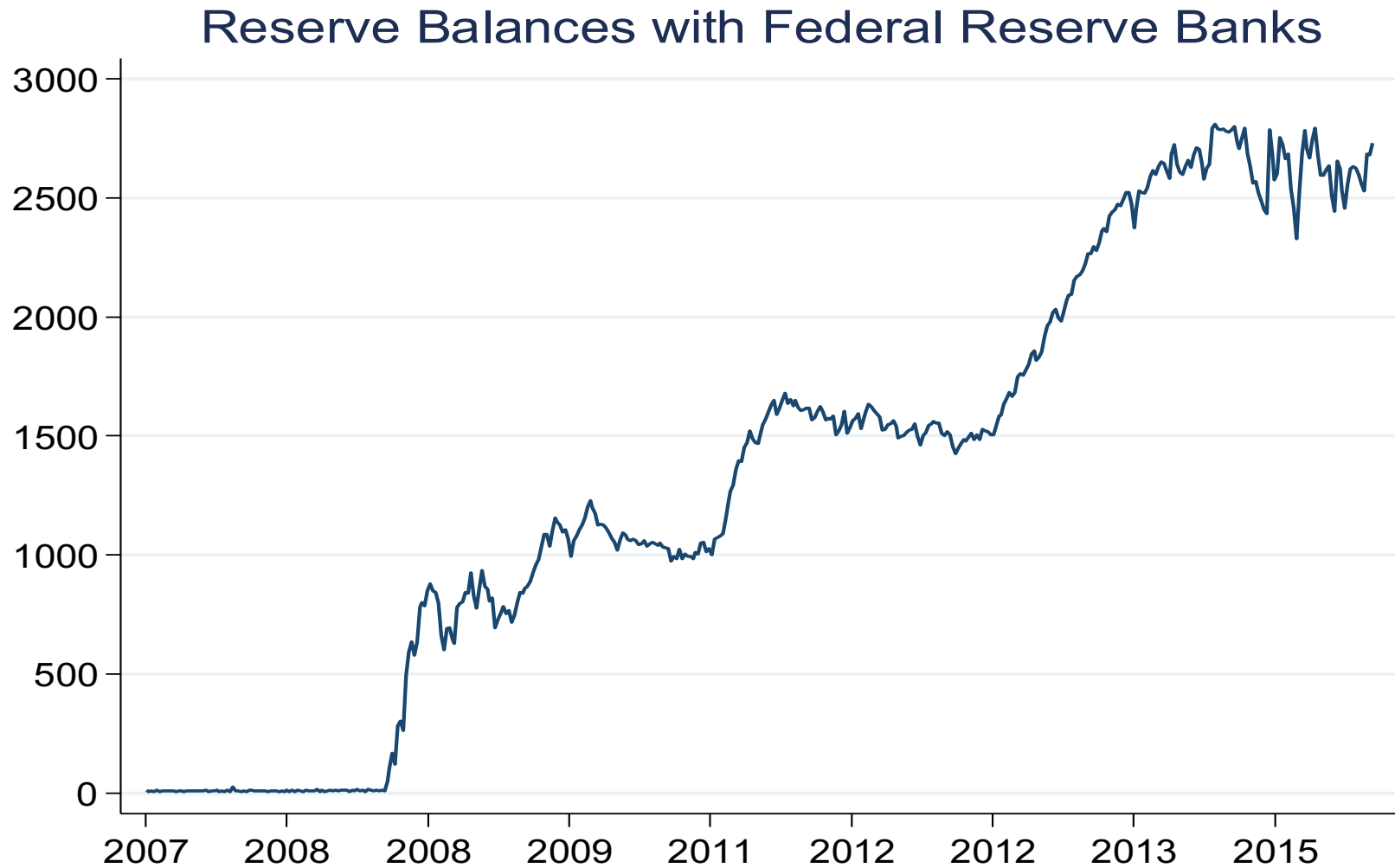


Pre-Crisis Operational Framework

- The level of reserve balances was quite low
 - Averaged around \$10 billion in 2006
- Desk and Board staff forecast factors driving supply of and demand for reserves
- Almost every day, conduct a repo operation to add or subtract enough reserves to hit the fed funds target rate

Monetary policy implementation since 2008

The crisis lead to a huge increase in reserves



First things first...

- Why are banks holding excess reserves?
 - Short answer: Because they have no choice!
- Reserves are injected in the system when the Federal Reserve purchases assets
- Reserves must be held by institutions that have a Fed account
- Banks can redistribute the reserves among themselves but cannot change the aggregate amount
 - Except for turning reserves into currency

Before 2008: Low amount of reserves

- Suppose there are two banks, A and B
- By assumption: reserves = required reserves
- Balance sheets in normal times (pre-crisis):

Bank A				Bank B			
Reserves	10	Deposits	100	Reserves	10	Deposits	100
Loans	50			Loans	130	Due to	
Due from						Bank A	40
Bank B	40						
Securities	10	Capital	10	Securities	10	Capital	10

- Interbank lending promotes efficient allocation of resources

The “crisis”

- Suppose Bank A is no longer willing to lend to Bank B and fails to roll over its overnight financing
 - May have concerns about credit risk or about its own funding needs
 - Bank B must now scramble to replace this financing so it can meet its required reserve ratios.

Bank A				Bank B			
Reserves	50	Deposits	100	Reserves	-30	Deposits	100
Loans	50			Loans	130	Due to Bank A	0
Due from Bank B	0					Capital	10
Securities	10	Capital	10	Securities	10		

The “crisis”

- Bank B must call in \$40 of loans, to increase its reserves
- The loans must show up as deposits somewhere
 - For simplicity, at Bank A
- Total reserves in the system stays fixed at 20

Bank A			
Reserves	10	Deposits	60
Loans	50		
Due from Bank B	0		
Securities	10	Capital	10

Bank B			
Reserves	10	Deposits	100
Loans	90	Due to Bank A	0
Securities	10	Capital	10

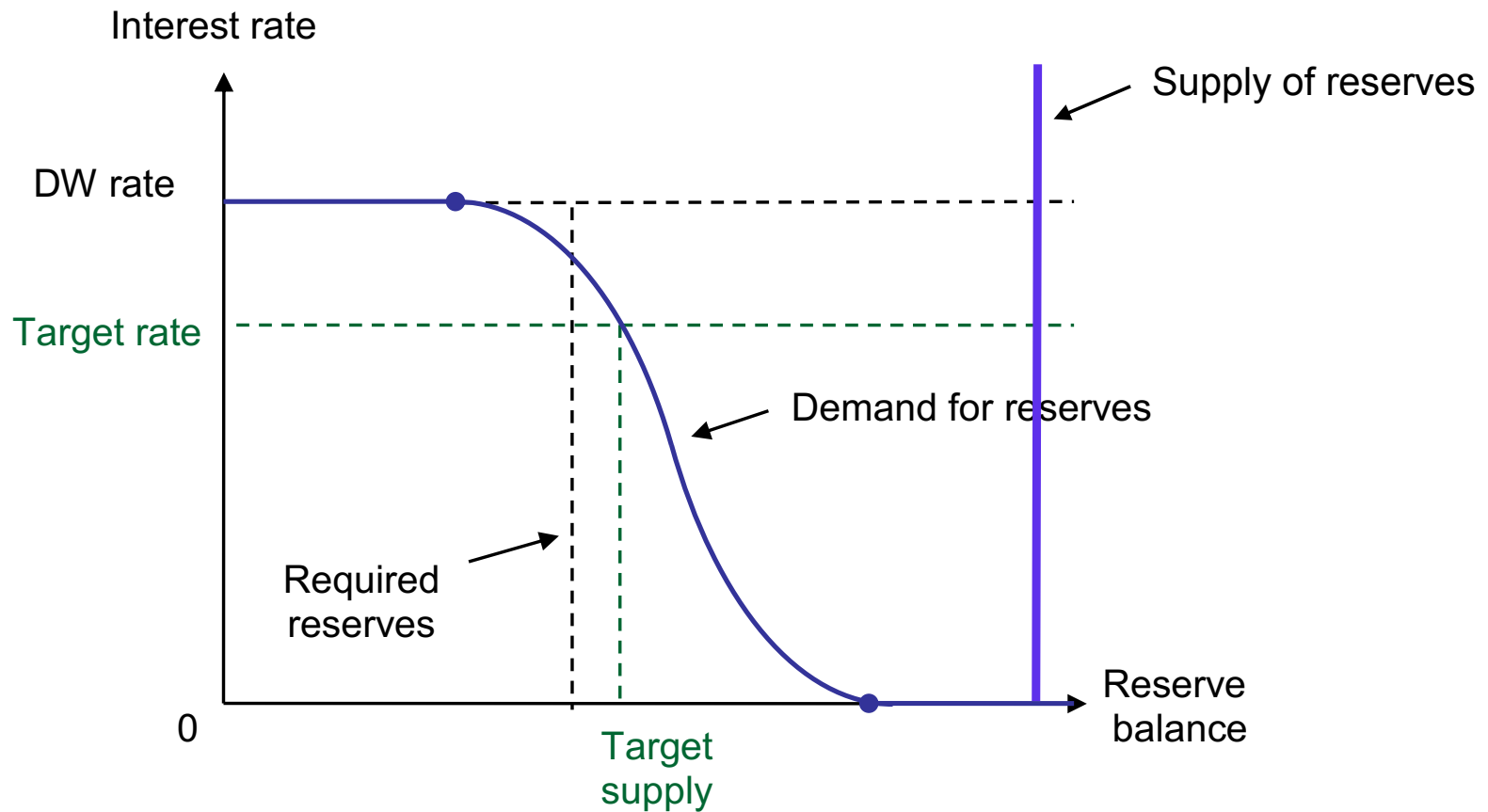
Central bank response to the crisis

- Suppose the central bank lends to Bank B
 - The decrease in lending can be avoided
 - This is the “lender of last resort” function of the central bank

Bank A				Bank B			
Reserves	50	Deposits	100	Reserves	10	Deposits	100
Loans	50			Loans	130	Due to CB	40
Securities	10	Capital	10	Securities	10	Capital	10

- Note: total reserves are now \$60, \$40 of which are excess
- The central bank’s policy is highly effective, even though it generates a large amount of excess reserves

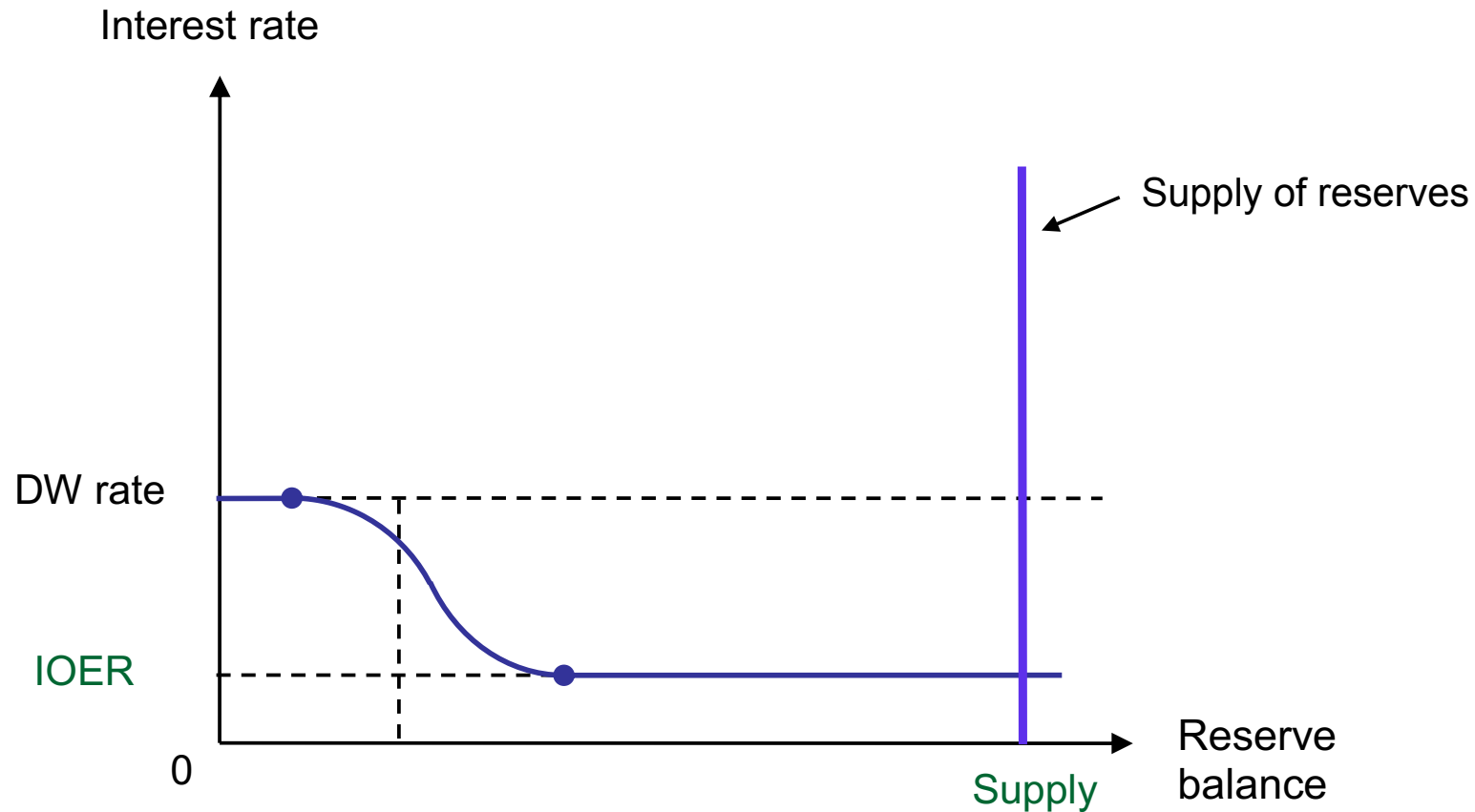
Equilibrium price of reserves is zero



Interest on excess reserves (IOER)

- Relatively recent authority granted to the Fed in 2008
- Allows the Fed to pay interest to banks on the balances that they leave in their account with the Fed
- How does it work?
 - If banks can earn, say, 0.25% leaving money in their Fed account, they won't have an incentive to lend it out below that rate
 - Should provide a floor on interbank rates

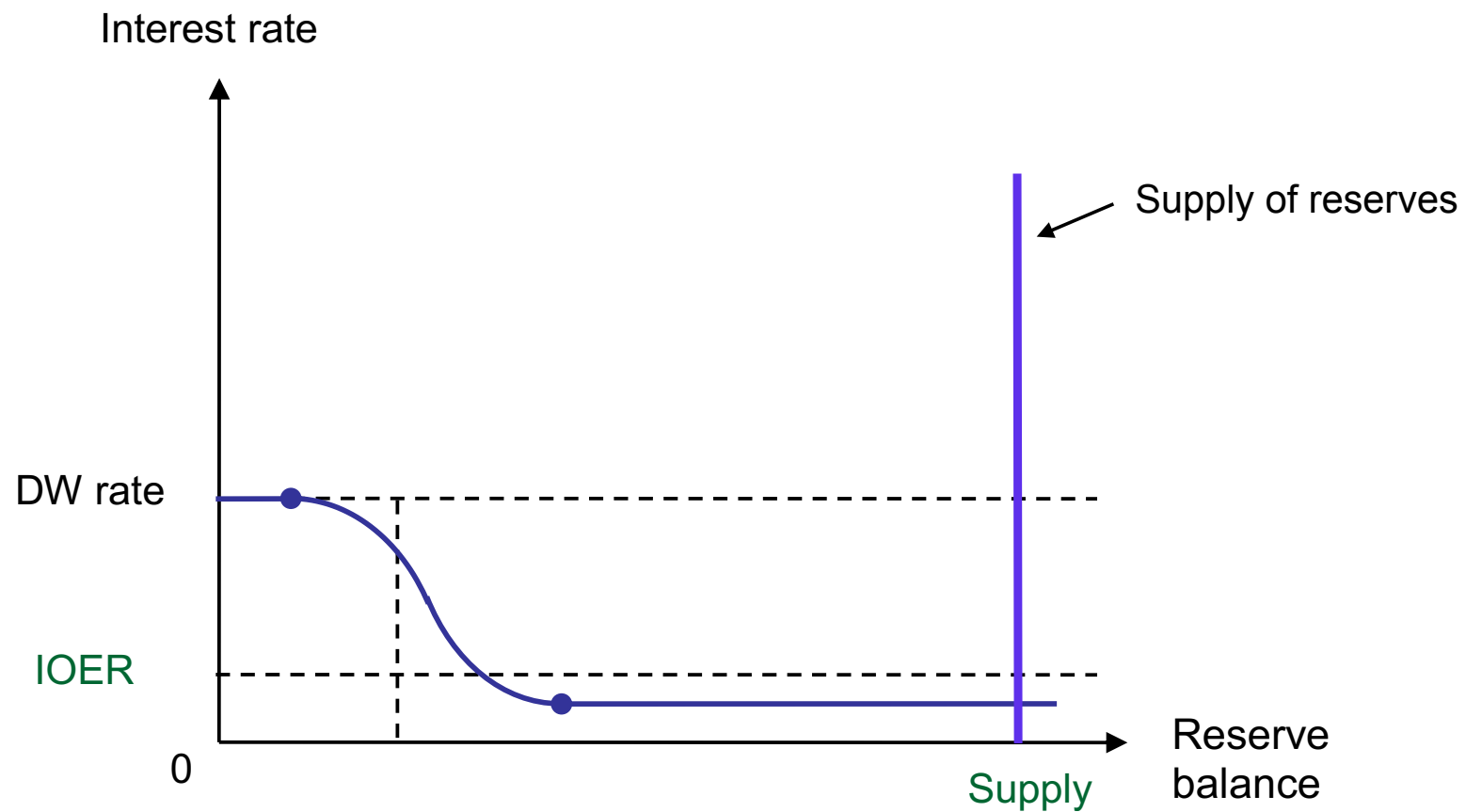
Monetary policy implementation since 2008



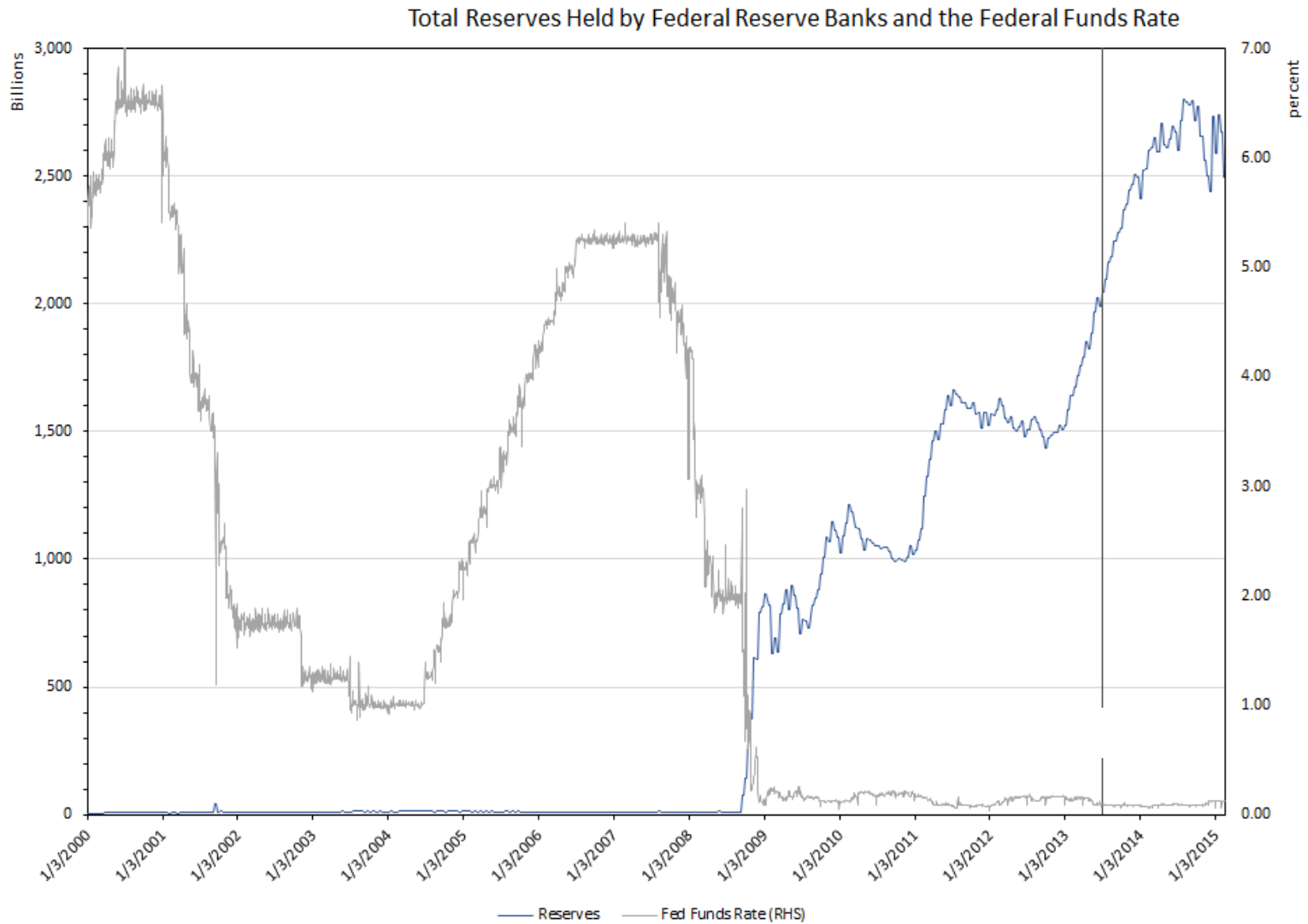
Implications

- Little need to forecast factors driving supply of and demand for reserves each day
- No need to conduct a repo operation to set the supply of reserves to the desired level
 - Small-scale adjustments to the supply of reserves have little to no impact on the fed funds rate

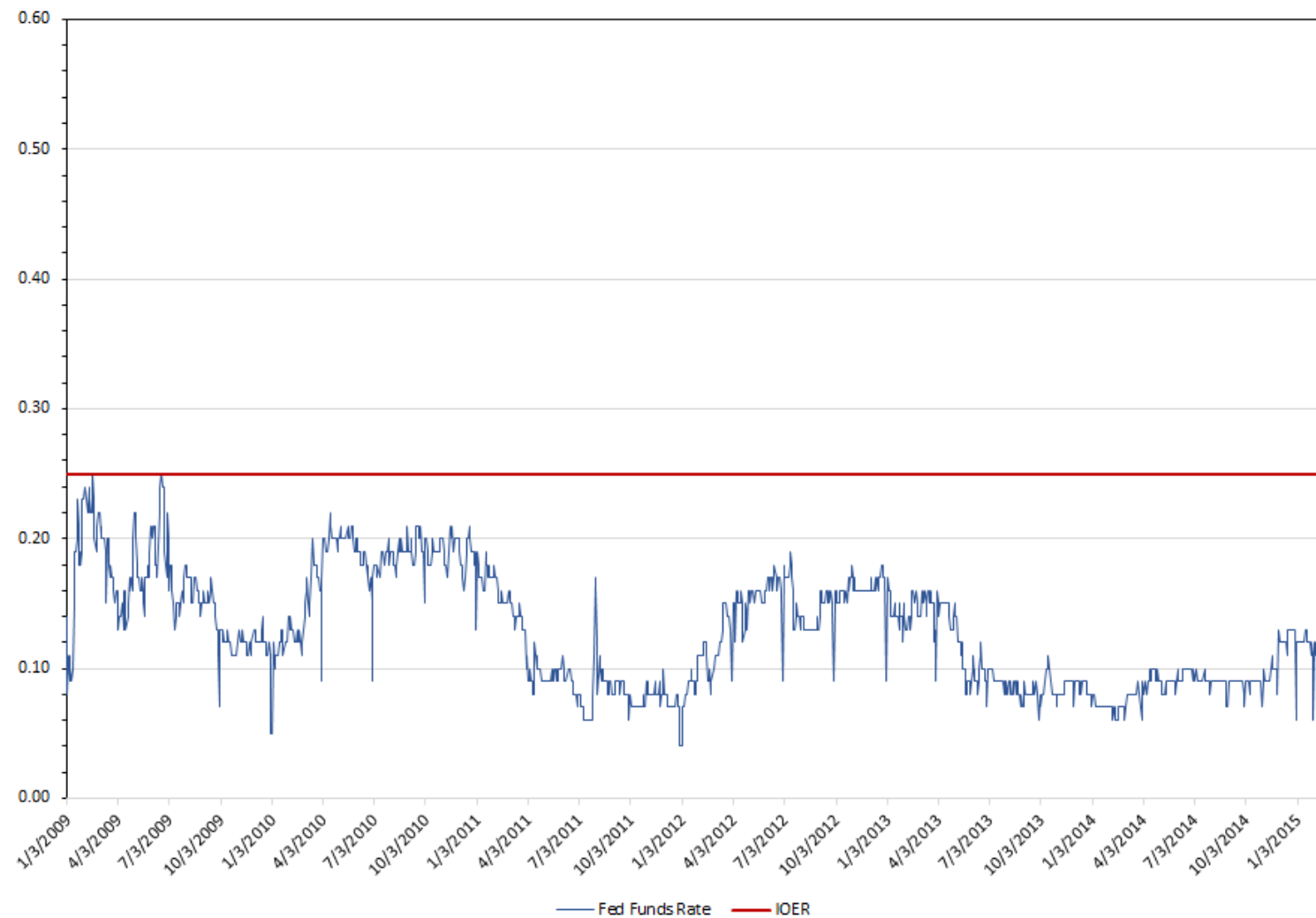
The “leaky floor”



Reserves and FFR



IOER and FFR



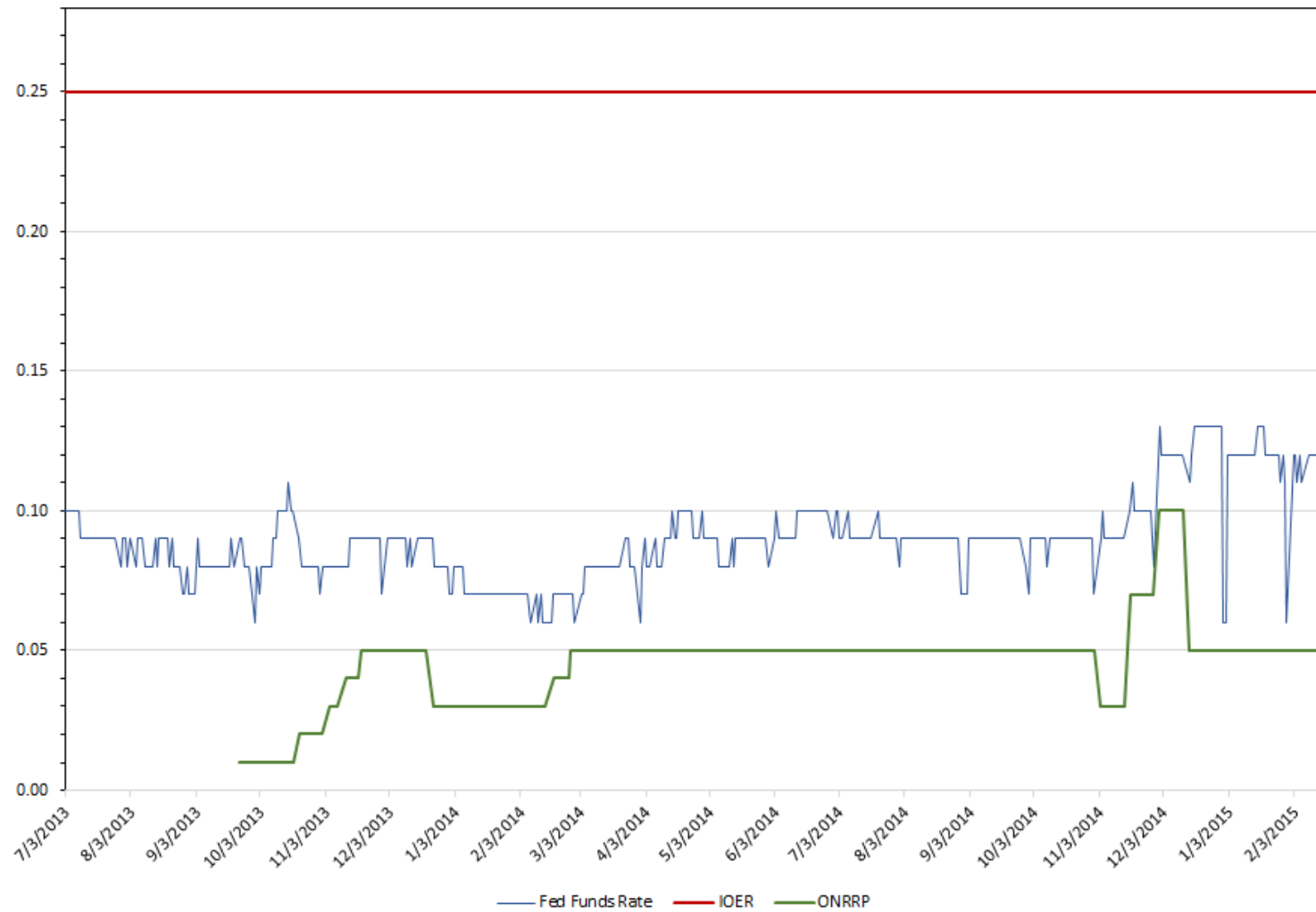
Why are market rates below the IOER?

- GSEs and FHLBs have Fed accounts, but cannot earn interest on reserves
- Two frictions appear to limit the pull of IOER:
 - Banks face balance sheet costs, related to new Basel III regulations
 - U.S. money market appears to be imperfectly competitive

ON RRP

- In August 2013, the FOMC announced that an overnight ON RRP facility could be used as another potential tool.
 - Fed borrows money from broad group of counter parties, bidders offer rates
 - Uniform (Single) Price Auction
 - Price is a “policy rate” if not oversubscribed
- ON RRP creates a new interest rate floor that is directly relevant to small set of non-banks.

IOER, FFR and ON RRP



Lift-off

